Week 4 | Solve With TAs

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Q1) Let the probability density function is defined as

$$f(x|\ \lambda) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k},\ \ x \geqslant 0, \lambda, k > 0$$

where k is some constant and λ is the parameter that we want to estimate using Maximum Likelihood Estimation. Choose the option which has the correct MLE form of λ.

- ○ $\left(\sum_{i=1}^{n} x_i^k\right)^{\frac{1}{k}}$
- ○ $\left(\frac{1}{n}\sum_{i=1}^{n} x_i^k\right)^{\frac{1}{k}}$
- ○ $\left(\frac{1}{n}\sum_{i=1}^{n} x_i^k\right)^{k}$
- ○ $\left(\frac{1}{n}\sum_{i=1}^{n} x_i^k\right)$

Solution:

Given the density function

$$f(x|\ \lambda) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k},\ \ x \geqslant 0$$

The likelihood function is

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^{n} f(x_i|\ \lambda) \\ &= \prod_{i=1}^{n} \frac{k}{\lambda}\left(\frac{x_i}{\lambda}\right)^{k-1} e^{-(x_i/\lambda)^k} \end{aligned}$$

The log likelihood function is (natural log):

$$l(\lambda) = \log(L(\lambda)) = \sum_{i=1}^{n}\left[\log\left(\frac{k}{\lambda}\right) + (k-1)\log\left(\frac{x_i}{\lambda}\right) - \left(\frac{x_i}{\lambda}\right)^k\right]$$

On simplifying

$$l(\lambda) = \sum_{i=1}^{n} \left[\log\left(\frac{k}{\lambda}\right) + (k-1)\log\left(\frac{x_i}{\lambda}\right) - \left(\frac{x_i}{\lambda}\right)^k \right]$$

$$= \sum_{i=1}^{n} \left[\log(k) - \log(\lambda) + (k-1)(\log(x_i) - \log(\lambda)) - \left(\frac{x_i}{\lambda}\right)^k \right]$$

$$= n\log(k) \cancel{- n\log(\lambda)} + (k-1)\sum_{i=1}^{n} \log(x_i) - kn\log(\lambda) + \cancel{n\log(\lambda)} - \sum_{i=1}^{n} \left(\frac{x_i}{\lambda}\right)^k$$

$$= n\log(k) - kn\log(\lambda) + (k-1)\sum_{i=1}^{n} \log(x_i) - \sum_{i=1}^{n} \left(\frac{x_i}{\lambda}\right)^k$$

$$l(\lambda) = -kn\log(\lambda) - \sum_{i=1}^{n} \left(\frac{x_i}{\lambda}\right)^k$$

We will ignore the terms that does not contain λ

Derivative with respect to λ

$$\frac{dl(\lambda)}{d\lambda} = \frac{d}{d\lambda}\left(-kn\log(\lambda) - \sum_{i=1}^{n}\left(\frac{x_i}{\lambda}\right)^k\right)$$

$$= -\frac{kn}{\lambda} + \frac{k}{\lambda^{k+1}}\sum_{i=1}^{n} x_i^k$$

- The derivative of $-kn\log(\lambda)$ is $-\dfrac{kn}{\lambda}$
- The derivative of $-\displaystyle\sum_{i=1}^{n}\left(\frac{x_i}{\lambda}\right)^k$ is $\dfrac{k}{\lambda^{k+1}}\displaystyle\sum_{i=1}^{n} x_i^k$

$$\frac{dl(\lambda)}{d\lambda} = -\frac{kn}{\lambda} + \frac{k}{\lambda^{k+1}}\sum_{i=1}^{n} x_i^k$$

Setting the derivative to zero

$$-\frac{kn}{\lambda} + \frac{k}{\lambda^{k+1}} \sum_{i=1}^{n} x_i^k = 0$$

$$k\left(-\frac{n}{\lambda} + \frac{1}{\lambda^{k+1}} \sum_{i=1}^{n} x_i^k\right) = 0$$

$$-\frac{n}{\lambda} + \frac{1}{\lambda^{k+1}} \sum_{i=1}^{n} x_i^k = 0$$

Multiply through by λ^{k+1}

$$-n\lambda^k + \sum_{i=1}^{n} x_i^k = 0$$

$$\lambda^k = \frac{1}{n}\sum_{i=1}^{n} x_i^k$$

Finally, the MLE for λ is

$$\widehat{\lambda}_{ML} = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^k\right)^{\frac{1}{k}}$$

Q2) Suppose that Y follows a binomial distribution with parameters n and $p = \theta$, so that the p.m.f. of Y given θ is:

$$P(Y;\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$$

for $y = 0, 1, \cdots, n$. Suppose that the prior p.d.f of the parameter θ is the beta(α, β) p.d.f., that is :

$$h(\theta) = \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

for $0 \leqslant \theta \leqslant 1$. Find the posterior p.d.f. of θ using Bayesian estimation that is $P(\theta; Y)$

Solution:

From Bayesian Estimation, we know that

$$\text{Posterior} \;\propto\; \text{Prior} \;\times\; \text{Likelihood}$$

Substitute the expressions for $P(Y;\theta)$ and $h(\theta)$ to get $P(\theta;Y)$

$$\begin{aligned} P(\theta;Y) &\propto P(Y;\theta) \times h(\theta) \\ &\propto \left(\binom{n}{y}\theta^{y}(1-\theta)^{n-y}\right) \times \left(\theta^{\alpha-1}(1-\theta)^{\beta-1}\right) \\ &\propto \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1} \end{aligned}$$

So the posterior p.d.f. is $\mathrm{Beta}(\alpha+y, \beta+n-y)$.

Q3) Consider a $1D$ dataset as $X = [-3, -1, 1, 3]$ with $K = 3$ mixture of Gaussian. Using EM algorithm, we try to estimate the mean and variance of the Gaussian component. Suppose the output of E-step in an iteration is

$$R_{n\times k} = \begin{array}{c} \xrightarrow{\quad k \quad} \\ \begin{bmatrix} 0.3 & a & 0.5 \\ b & 1 & b \\ c & 0.5 & c \\ 0.1 & 0.4 & d \end{bmatrix} \end{array} \Bigg\downarrow n$$

where each R_{ij} entry corresponds to the probability of X_i data points belongs to component K_j that is λ_j^i. Let $\widehat{\mu_1}, \widehat{\mu_2}$ and $\widehat{\mu_3}$ be the updated mean in M-step after E-step. Calculate $\widehat{\mu_1} + \widehat{\mu_2} + \widehat{\mu_3}$.

Solution:

First we will fill the missing values in the matrix R using the fact that $\sum_{k=1}^{K} \lambda_k^i = 1$. The updated matrix R is

$$R_{n\times k} = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0 & 1 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

Now the updated mean in M-step is the weighted average that is $\mu_k = \dfrac{\sum_{i=1}^{n} \lambda_k^i \cdot x_i}{\sum_{i=1}^{n} \lambda_k^i}$

$$\widehat{\mu_1} = \frac{(0.3 \times (-3)) \ + \ (0 \ \times (-1)) \ + \ (0.25 \ \times 1) \ + (0.1 \times 3)}{0.3 + 0 + 0.25 + 0.1}$$

$$= \frac{-0.35}{0.65}$$

$$= -0.53$$

$$\widehat{\mu_2} = \frac{(0.2 \times (-3)) \ + \ (1 \times (-1)) \ + \ (0.5 \ \times 1) \ + (0.4 \times 3)}{0.2 + 1 + 0.5 + 0.4}$$

$$= \frac{0.1}{2.1}$$

$$= 0.04$$

$$\widehat{\mu_3} = \frac{(0.5 \times (-3)) \ + \ (0 \times (-1)) \ + \ (0.25 \ \times 1) \ + (0.5 \times 3)}{0.5 + 0 + 0.25 + 0.5}$$

$$= \frac{0.25}{1.25}$$

$$= 0.2$$

Final Ans: $\widehat{\mu_1} + \widehat{\mu_2} + \widehat{\mu_3} = \ -0.53 + 0.04 + 0.2 = \ -0.29$

Q4) A GMM is used to model a dataset with 3 components. For a given data point $x = 6$, the likelihoods for each component are as follows:

$$P(x \mid z = 1) = 0.2$$
$$P(x \mid z = 2) = 0.3$$
$$P(x \mid z = 3) = 0.5$$

The mixture probabilities for the components are $\pi_1 = 0.4,\ \pi_2 = 0.3,\ \pi_3 = 0.3$. Which component is most likely to have generated the data point?

Hint: Use posterior probability

Solution: You are given a GMM with 3 components, and the following likelihoods and weights are given for a data point $x = 6$

$$P(x \mid z = 1) = 0.2$$
$$P(x \mid z = 2) = 0.3$$
$$P(x \mid z = 3) = 0.5$$

Mixture Probabilities

$$\pi_1 = 0.4$$
$$\pi_2 = 0.3$$
$$\pi_3 = 0.3$$

Step 1: Calculate the posterior probability of each component for the data point $x = 6$ which is given by

$$P(z = k \mid x) = \frac{\pi_k \cdot P(x \mid z = k)}{\sum_{k=1}^{K} \pi_k \cdot P(x \mid z = k)}$$

- Let us calculate the numerator for each component

  - Component 1: $\pi_1 \cdot P(x \mid z = 1) = 0.4 \ \times 0.2 = 0.08$
  - Component 2: $\pi_2 \cdot P(x \mid z = 2) = 0.3 \ \times 0.3 = 0.09$
  - Component 3: $\pi_3 \cdot P(x \mid z = 3) = 0.3 \ \times 0.5 = 0.15$

- Denominator (sum of the numerators)

  - $0.08 + 0.09 + 0.15 = 0.32$

Now, we compute the posterior probability for each component:

$$P(z = 1 \mid x) = \frac{0.08}{0.32} = 0.25 \qquad P(z = 2 \mid x) = \frac{0.09}{0.32} = 0.28125 \qquad P(z = 3 \mid x) = \frac{0.15}{0.32} = 0.46875$$

The component with the highest posterior probability is the most likely to have generated the data point $x = 6$. Thus, Component 3 is the most likely to have generated the data point.

Q6) A dataset has n natural numbers. If we assume a Poisson distribution with parameter λ as the probabilistic model.

1. What is the likelihood function
2. What is the maximum likelihood estimate for λ

Hint: P.M.F of Poisson distribution is given by :

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

Solution:

Likelihood function is given by :

$$
\begin{aligned}
L(\lambda, \{x_1,..., x_n\}) &= \prod_{i=1}^{n} P(x_i|\lambda) \\
&= \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} \\
&= \frac{e^{-n\lambda}\lambda^{\sum_{i=1}^{n} x_i}}{\prod_{i=1}^{n} x_i!}
\end{aligned}
$$

On taking $\ln$, we get :

$$= \ln\left(e^{-n\lambda}\right) + \ln\left(\lambda^{\sum_{i=1}^{n} x_i}\right) - \ln\left(\prod_{i=1}^{n} x_i!\right)$$

$$= -n\lambda + \sum_{i=1}^{n} x_i \ln(\lambda) - \ln\left(\prod_{i=1}^{n} x_i!\right)$$

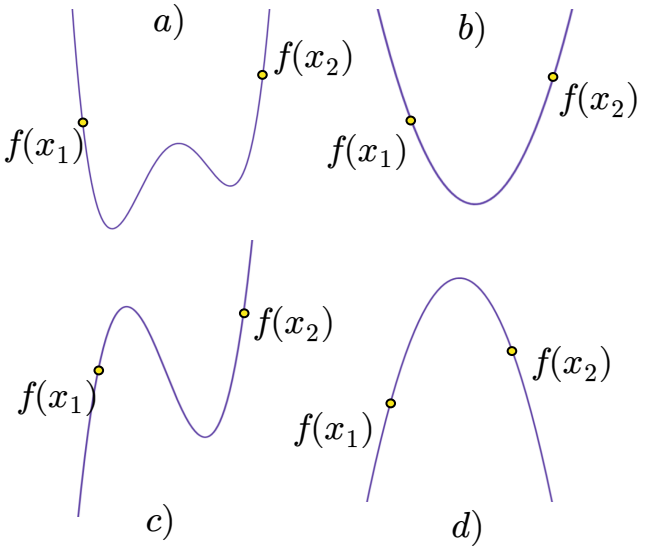
Differentiate with respect to λ as equate to 0

$$-n + \frac{\sum_{i=1}^{n} x_i}{\lambda} = 0$$

$$-n = \frac{-1}{\lambda}\sum_{i=1}^{n} x_i$$

$$\lambda = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Q6) Look at the following graphs.



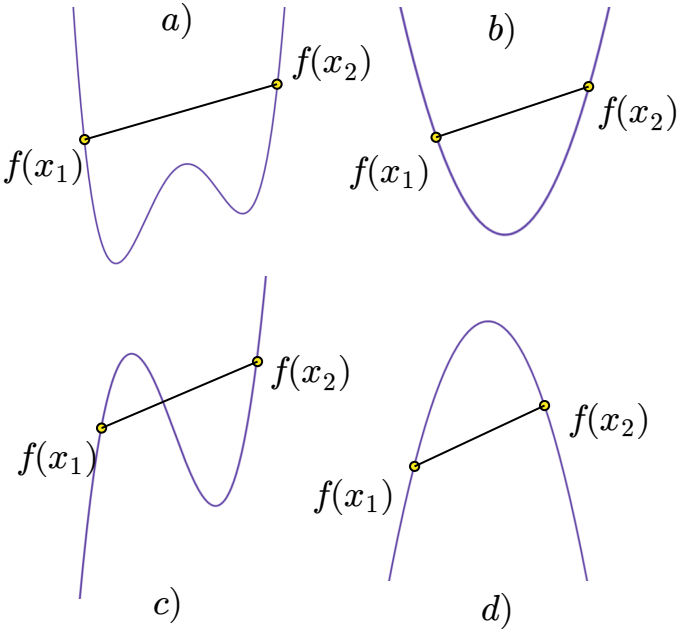
1. Select the option(s) which satisfy the following inequality for $\lambda_1, \lambda_2 \in [0, 1]$ and $\lambda_1 + \lambda_2 = 1$,

$$f(\lambda x_1 + \lambda_2 x_2) \leqslant \lambda_1 f(x_1) + \lambda_2 f(x_2)$$

2. Select the graph(s) from above that represent convex functions.

Solution:

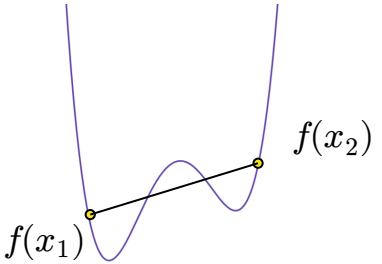
Let us linearly interpolate $f(x_1)$ and $f(x_2)$



We can see that the first two options satisfy the given inequality.

2) The last two options cannot be convex functions as they didn't satisfy the inequality in option 1.

The function in $a)$ is not a convex function. On choosing a different x_1 and x_2, we get the following.



The inequality in option 1 is not satisfied for the new x_1 and x_2.

For a function to be a convex function, the inequality in option 1 should be satisfied for all x_1 and x_2 in the domain of the function.

Q7) We are trying to fit a GMM $(K = 3)$ for the following data points.

$$\{2, 1, 3, 2, 1, 2, 0, -3, 0, 2\}$$

At the tth round of the EM algorithm, the following were observed.

$$\lambda_3 = (\,0.2,\quad 0.3,\quad 0.1,\quad 0.6,\quad 0.1,\quad 0.5,\quad 0.4,\quad 0.7,\quad 0.1,\quad 0.2\,)$$

The ith element of this vector represents $P(z_i = 3|x_i)$. What will μ_3 and π_3 be after the maximization step in the tth round.

Solution:

The ith element of λ_3 represents $P(z_i = 3|x_i)$. From the notations used in the lectures, it is λ_3^i. That is the probability that given data point x_i, it belongs to component $z_i = 3$

μ_3 and π_3 are given by the following :

$$\mu_3 = \frac{\sum_{i=1}^{n} \left(\lambda_3^i\right)(x_i)}{\sum_{i=1}^{n} \lambda_3^i} \qquad \pi_3 = \frac{\sum_{i=1}^{n} \lambda_3^i}{n}$$

$n = 10.$

$$\sum_{i=1}^{10} \lambda_3^i = 0.2 + 0.3 + 0.1 + 0.6 + 0.1 + 0.5 + 0.4 + 0.7 + 0.1 + 0.2 = 3.2$$

$$\sum_{i=1}^{n}\left(\lambda_3^i\right)(x_i) = (0.2\times 2) + (0.3\times 1) + (0.1\times 3) + (0.6\times 2) + (0.1\times 1) + (0.5\times 2) + (0.4\times (0)) + (0.7\times (-3)) + (0.1\times 0) + (0.2\times 2)$$

$$\sum_{i=1}^{n}\left(\lambda_3^i\right)(x_i) = 0.4 + 0.3 + 0.3 + 1.2 + 0.1 + 1 - 2.1 + 0.4$$

$$\sum_{i=1}^{n}\left(\lambda_3^i\right)(x_i) = 1.6$$

Therefore,

$$\mu_3 = \frac{\sum_{i=1}^{n}\left(\lambda_3^i\right)(x_i)}{\sum_{i=1}^{n}\lambda_3^i} = \frac{1.6}{3.2} = 0.5$$

$$\pi_3 = \frac{\sum_{i=1}^{n}\lambda_3^i}{n} = \frac{3.2}{10} = 0.32$$

Q8) We have n iid samples from exponential distribution with λ as parameter. λ is said to follow a gamma distribution with α and β as parameters. Find the posterior distribution.

Hint: The PDF of exponential distribution with parameter λ is given by $f(x) = \lambda e^{-\lambda x}$.

The PDF of gamma distribution $f(\lambda) = \dfrac{\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$

Find the posterior mean for $\alpha = 6$ and $\beta = 2$. The data points are $\{2, 1, 1, 0.75, 1.25, 1\}$.

Solution: The likelihood function is given by :

$$\begin{aligned} L(\lambda, \{x_1,..., x_n\}) &= \prod_{i=1}^{n} \lambda e^{-\lambda x_i} \\ &= \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i} \end{aligned}$$

Prior is given by $f(\lambda) = \dfrac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$. Here, $\dfrac{\beta^\alpha}{\Gamma(\alpha)}$ can be treated as a constant.

We know that the posterior is proportional to Likelihood $\times$ Prior. This is given by :

$$\begin{aligned} &= \left(\lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}\right) \times \left(\lambda^{\alpha-1} e^{-\beta\lambda}\right) \\ &= \left(\lambda^{n+\alpha-1}\right) \left(e^{-\lambda\left(\beta + \sum_{i=1}^{n} x_i\right)}\right) \end{aligned}$$

The posterior distribution is $\mathrm{Gamma}\left(n+\alpha, \beta + \sum_{i=1}^{n} x_i\right)$.

We have 6 data points. Therefore, $n + \alpha = 6 + 6 = 12$.

$$\sum_{i=1}^{6} x_i = 2 + 1 + 1 + 0.75 + 1.25 + 1 = 2$$

Therefore, $\beta + \sum_{i=1}^{n} x_i = 2 + 7 = 9$.

The mean of Gamma distribution is given by $\frac{\alpha}{\beta}$.

The posterior mean is $\frac{12}{9} = 1.333$